PROJECT PROPOSAL

Project Title: Hotel Booking Performance, Cancellation and Revenue Analysis Using Power BI

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GitHub Repository: To be added

Problem Statement

Hotel booking data contains important information about booking volumes, cancellations, revenue, customer behaviour, lead time, deposit type, market segment, and hotel performance. Traditional reporting can make it difficult to identify patterns and compare performance across different booking situations. This project aims to develop an interactive, data-driven Power BI dashboard that analyses hotel booking performance, identifies factors associated with cancellations, and provides insights into revenue and booking trends for City Hotel and Resort Hotel.

Methodology

The project will use a cleaned hotel booking dataset containing booking and customer-related attributes. The data will first be cleaned and transformed using Power Query, followed by exploratory analysis to identify patterns in bookings, cancellations, revenue, ADR, lead time, deposit type, market segment, hotel type, and monthly performance. DAX measures will then be created to calculate key indicators such as total bookings, completed bookings, cancelled bookings, cancellation rate, realised revenue, average ADR, and average stay. The final results will be presented through interactive Power BI visualisations, KPI cards, hotel filters, and analytical charts.

Expected Outcome

The project will produce an interactive Power BI dashboard for analysing hotel booking performance and cancellation behaviour. It will provide data-driven insights into cancellation rates by lead time, deposit type, and market segment, together with comparisons of revenue, ADR, completed bookings, and cancelled bookings between City Hotel and Resort Hotel. The project will demonstrate how data preparation, DAX-based analysis, business intelligence, and interactive visualisation can be applied to support hotel revenue management and decision-making.